Andrew Brettin

New York, NY brettin@cims.nyu.edu Website GitHub Linkedin

Purpose

I'm a climate scientist specializing in machine learning and sea level. My interests center on the development and application of data-driven tools for assessing predictability and quantifying risks from sea level variability.

Education

PhD, Atmosphere-Ocean Science and Mathematics — **May 2025**
Courant Institute of Mathematical Sciences, New York University — New York, NY
Advisor: Laure Zanna

Bachelor of Science, Mathematics — **May 2019**
University of Minnesota, College of Science & Engineering — Minneapolis, MN
Summa cum laude with high distinction (GPA 3.92)

Research Projects

Estimating temperature and precipitation extremes using quantile neural networks | 2024–2025 [pdf] [code]

- Devised a novel yet straightforward quantile neural network framework for estimating nongaussian uncertainties
- Constrained estimates of tide gauge sea level observations under ERA5-estimated atmospheric conditions

Learning improved propagators for regional sea surface height dynamics | 2024 [publication] [code]

- Developed a machine learning architecture based on Koopman operator theory to learn an improved propagator for regional sea surface height forecasts
- Enhanced prediction skill by $\sim$5%–10% over conventional statistical-dynamical techniques (linear inverse models)

Identifying sources of sea level predictability with mean-variance networks | 2023–2024 [publication] [code]

- Leveraged uncertainty-quantifying neural networks to identify changes in sources of dynamic sea level predictability over daily-to-seasonal forecast leads in the CESM2 large ensemble simulation

Exploring nonstationarity of coastal sea level distributions | 2021–2022 [publication] [code]

- Collaborated on the development of a statistical theory for interpreting changes in probability distributions
- Analyzed trends in probability distributions of sea level in tide gauges and coupled climate models

Technical Skills

Domain knowledge	Climate dynamics • Machine Learning • Numerics • Probability • Dynamical Systems
Programming languages	Python (scipy, dask, xarray, scikit-learn, PyTorch, Keras) • MATLAB • Julia • C++
Workflows	Bash • git/GitHub • conda • VS Code • Jupyter • Pytorch Lightning • Weights & Biases
Computational skills	Numerical methods (optimization, quadrature, interpolation, finite difference, spectral methods) • MCMC • High performance computing • Distributed data parallelism
Statistics and ML	Linear/logistic regression • PCA • Maximum likelihood estimation • Unsupervised learning • Gaussian processes • Autoencoders • CNNs • Statistical-dynamical techniques (Linear inverse models, dynamic mode decomposition, Kalman filtering)

Communication Experience

- **Selected presentations**
  - **Andrew Brettin**, Laure Zanna, and Elizabeth Barnes (2023). *Identifying Drivers of Subseasonal-to-Seasonal Sea Level Predictability Using Uncertainty-Permitting Machine Learning*. Oral session, AGU Fall Meeting.
  - **Andrew Brettin** and Laure Zanna (2022). *Constraining Estimates for South American Sea Level Extremes Using Uncertainty-Permitting Machine Learning*. Poster session, AGU Fall Meeting.
  - **Andrew Brettin** and Laure Zanna (2022). *Characterizing the Impacts of Continental Shelf Depth on Sea Level Variability Using Clustering*. Poster session, AGU Ocean Sciences Meeting.
- **Teaching and tutoring**
  - Recitation leader, Numerical Methods, *New York University* (Fall 2022)
  - Peer tutor, Honors Calculus I–IV, *University of Minnesota, University Honors Program* (Fall 2016–Spring 2019)

Publications

1. **Brettin, A.** *Data-Driven Approaches for Predictability and Uncertainty Quantification of Sea Level Variability.* 3223742845. PhD thesis (New York University, 2025).
2. **Brettin, A.**, Zanna, L. & Barnes, E. A. Learning Propagators for Sea Surface Height Forecasts Using Koopman Autoencoders. *Geophysical Research Letters* **52.** 10.1029/2024GL112835, e2024GL112835 (2025).
3. **Brettin, A.**, Zanna, L. & Barnes, E. A. Uncertainty-Permitting Machine Learning Reveals Sources of Dynamic Sea Level Predictability across Daily to Seasonal Time Scales. *Artificial Intelligence for the Earth Systems* **4.** 10.1175/AIES-D-25-0014.1, 250014 (2025).
4. Falasca, F., **Brettin, A.**, Zanna, L., Griffies, S. M., Yin, J. & Zhao, M. Exploring the nonstationarity of coastal sea level probability distributions. *Environmental Data Science* **2.** 10.1017/eds.2023.10, e16 (2023).
5. Meyer, K., Broda, J., **Brettin, A.**, Sánchez Muñiz, M., Gorman, S., Isbell, F., Hobbie, S. E., Zeeman, M. L. & McGehee, R. Nitrogen-induced hysteresis in grassland biodiversity: a theoretical test of litter-mediated mechanisms. *The American Naturalist* **201.** 10.1086/724383, E153–E167 (2023).
6. **Brettin, A.**, Rossi-Goldthorpe, R., Weishaar, K. & Erovenko, I. V. Ebola could be eradicated through voluntary vaccination. *Royal Society Open Science* **5.** 10.1098/rsos.171591, 171591 (2018).

Workshops

- NASA/JPL Summer School on Satellite Observations and Climate Models | 2023
  Keck Institute for Space Studies, Caltech, Pasadena, CA
- LEAP Momentum Bootcamp on Climate Data Science | 2022
  Columbia University, New York, NY
- OceanHackWeek Data Science and Oceanography Interactive Workshop | 2021
  University of Washington eScience Institute, Virtual workshop
- Workshop on Climate Change and Resilience: Methods of Dynamical Systems and Data Assimilation | 2018
  American Institute of Mathematics, San Jose, CA

Service and Outreach

- Vice President, Courant Student Organization, *New York University* (Fall 2021–Summer 2022)
- Volunteer tutor, math grades 5–8, *Common Denominator* (Fall 2021–Spring 2022)
- Project mentor, Undergraduate Research Program in Data Science, *NYU Center for Data Science, in collaboration with the National Society of Black Physicists* (Spring 2021)
- Reviewer, *Geophysical Research Letters* (2025–) and *Journal of Geophysical Research: Machine Learning and Computation* (2025–)